Hope Fair Teaching Clinic

Breast Imaging · Ultrasound

PATIENT FOLDER

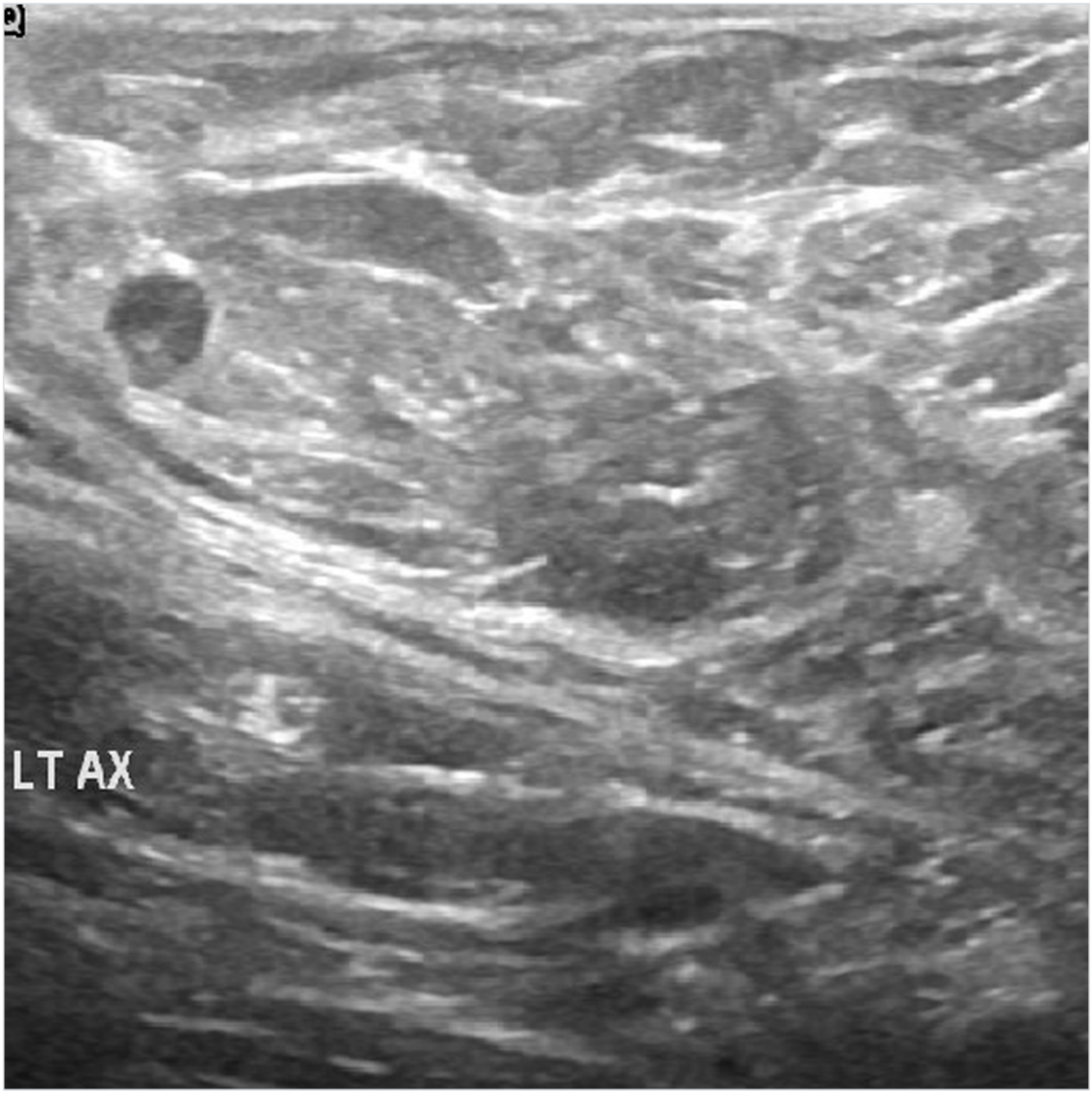
Folder 1

PATIENT	Mariam K.
AGE RANGE	45–50
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Follow-up of a lump noticed during a routine check

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

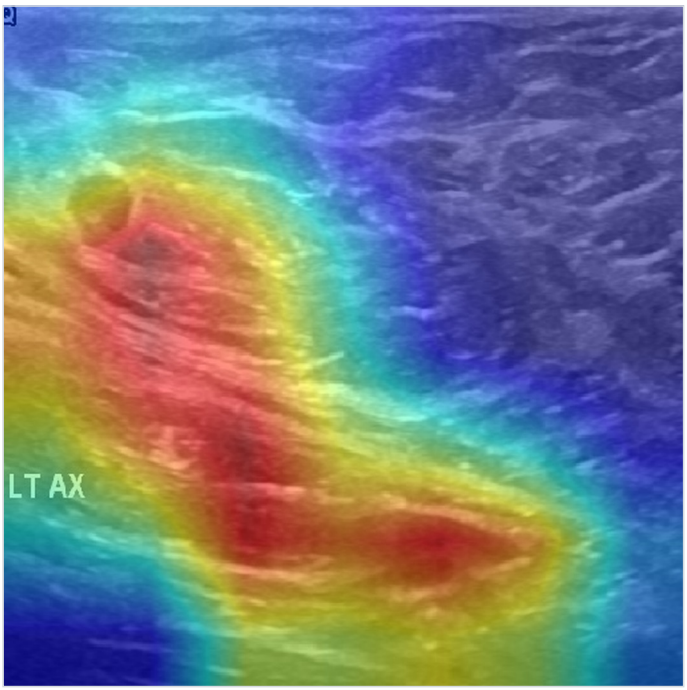


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Mariam K.	Examination	Breast ultrasound
Age range	45–50	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

BENIGN

MODEL CONFIDENCE PER CLASS

Normal	6%
Benign	**91%**
Malignant	3%

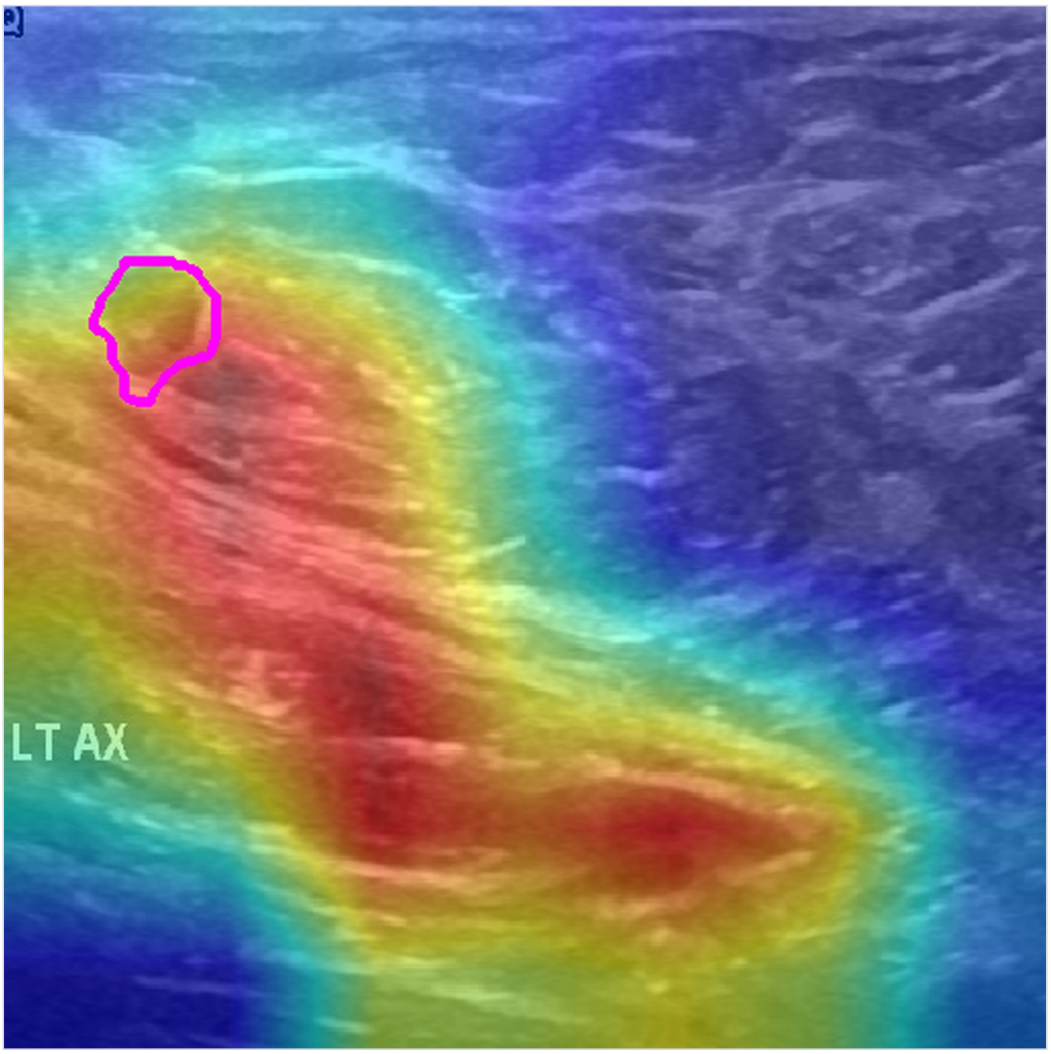
RECOMMENDATION

Likely benign. Routine follow-up suggested.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: MALIGNANT (cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The tumor is the small dark area near the top left. The AI's heatmap is a wide band through other tissue and only brushes the tumor's edge.

Verdict

Folder 1: Confidently wrong

FALSE NEGATIVE

نتيجة سلبية كاذبة

Arabic: draft, needs native review

The AI was 91% sure it was benign. It was cancer.

WHAT THIS COULD MEAN FOR THE PATIENT

- False reassurance: the patient is told not to worry.
- Diagnosis is delayed.
- In the meantime the cancer may grow or spread.
- It may be found later, when treatment is less likely to succeed.
- Confidence is not truth.

How a human doctor protects the patient

- Compares this scan with the patient's previous scans.
- A second radiologist reads the images independently.
- Any lump that worries the doctor goes to biopsy, whatever the software says.
- Looks at the whole patient: history, symptoms and examination.

Arabic: draft, needs native review

- يقارن هذه الصورة بصور المريضة السابقة.
- يقرأ طبيب أشعة ثان الصور بشكل مستقل.
- أي كتلة تقلق الطبيب تحال إلى الخزعة، مهما قال البرنامج.
- ينظر إلى المريضة ككل: تاريخها الصحي وأعراضها والفحص السريري.

QUESTION FOR YOU

The AI was 91% sure. What would make you trust, or doubt, a confident answer?

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Breast Imaging · Ultrasound

PATIENT FOLDER

Folder 2

PATIENT	Fatima A.
AGE RANGE	55–60
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Check of a breast change the patient noticed

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

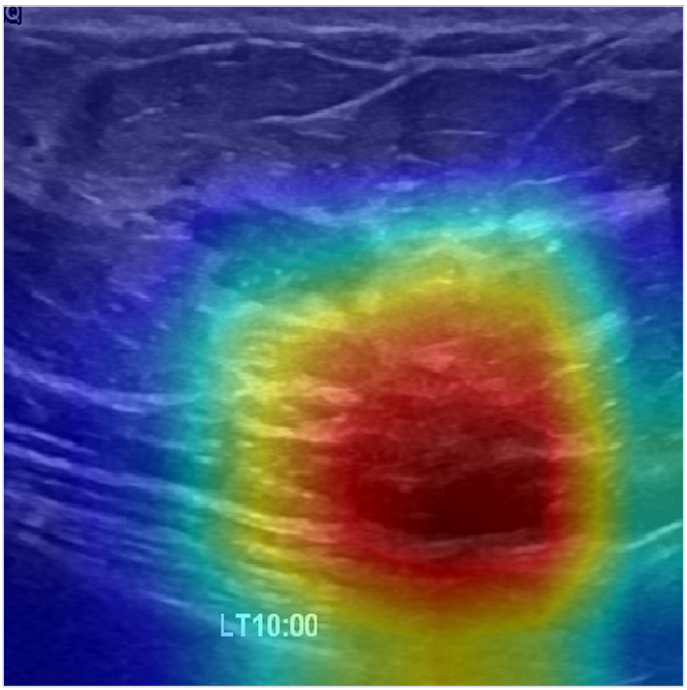


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Fatima A.	Examination	Breast ultrasound
Age range	55–60	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

BENIGN

MODEL CONFIDENCE PER CLASS

Normal 2%

Benign 90%

Malignant 8%

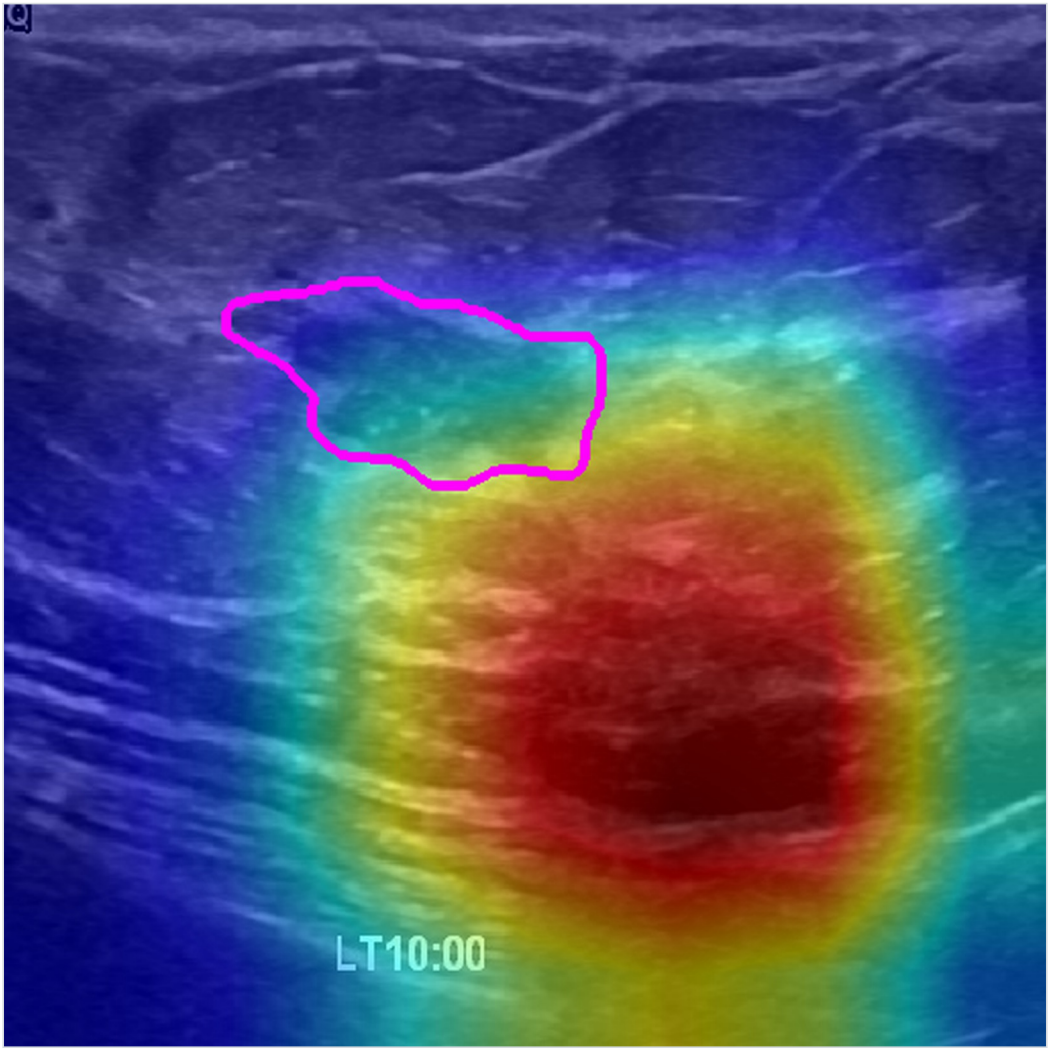
RECOMMENDATION

Likely benign. Routine follow-up suggested.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: MALIGNANT (cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The tumor is the dark area in the upper middle. The AI's heatmap is below and to the right of it, almost completely missing the tumor.

Verdict

Folder 2: Missed cancer

FALSE NEGATIVE

نتيجة سلبية كاذبة

Arabic: draft, needs native review

The AI looked in the wrong place and missed the cancer.

WHAT THIS COULD MEAN FOR THE PATIENT

- False reassurance: the patient goes home thinking all is well.
- Diagnosis is delayed.
- The cancer may grow or spread in the meantime.
- It may be found later, when treatment is less likely to succeed.

How a human doctor protects the patient

- Checks that the area of concern was actually looked at.
- A second reader reviews the images.
- Arranges follow-up imaging if anything is unclear.
- Orders a biopsy when in doubt.

Arabic: draft, needs native review

- يتأكد من أن المنطقة المثيرة للقلق قد فحصت فعلا.
- يراجع قارئ ثان الصور.
- يطلب تصويرا للمتابعة إذا كان هناك أي غموض.
- يطلب خزعة عند الشك.

QUESTION FOR YOU

The heatmap shows the AI looked in the wrong place. How could a doctor catch this mistake?

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Breast Imaging · Ultrasound

PATIENT FOLDER

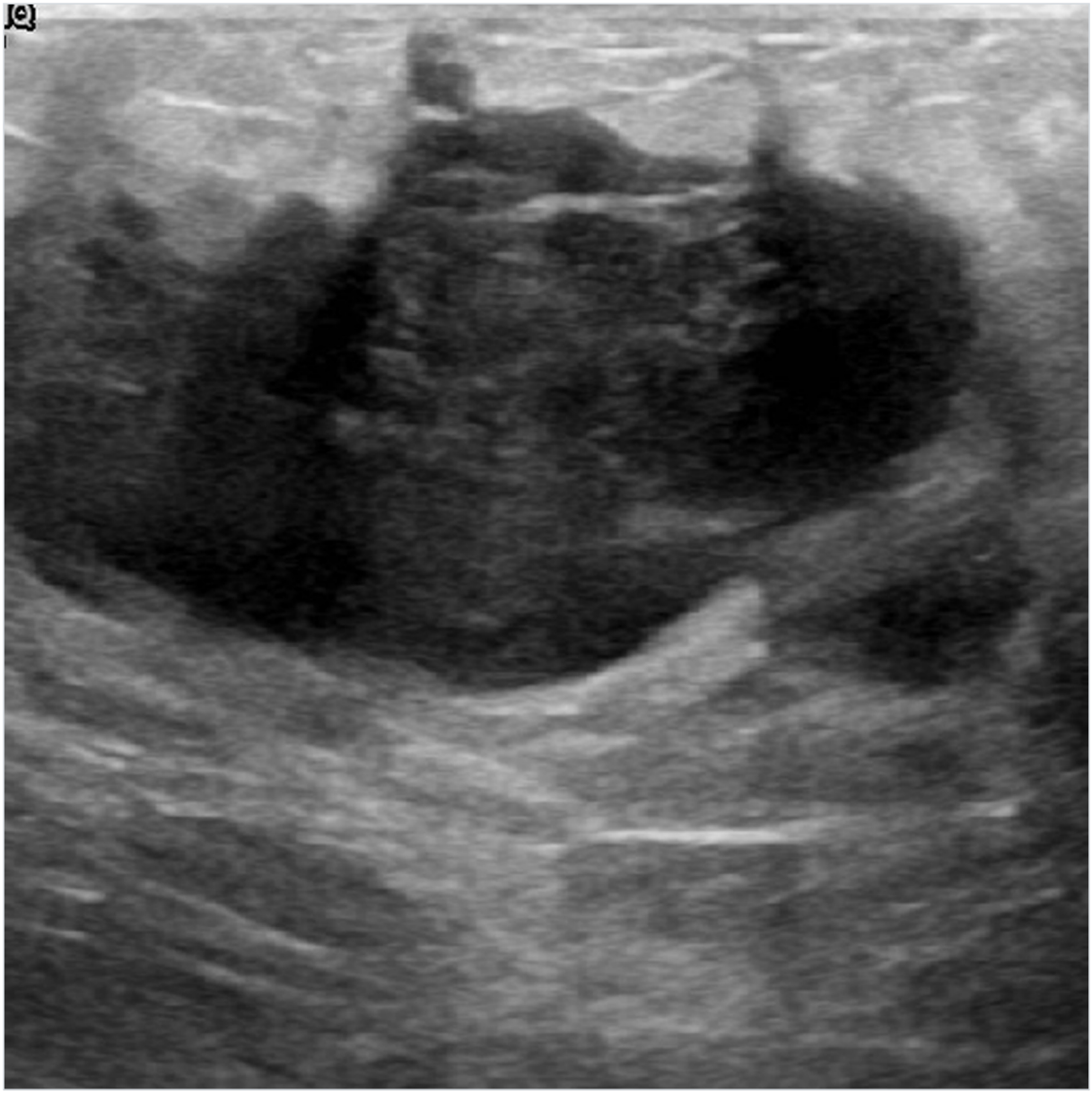
Folder 3

PATIENT	Noura S.
AGE RANGE	60–65
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Scan of a lump found during a clinical breast examination

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

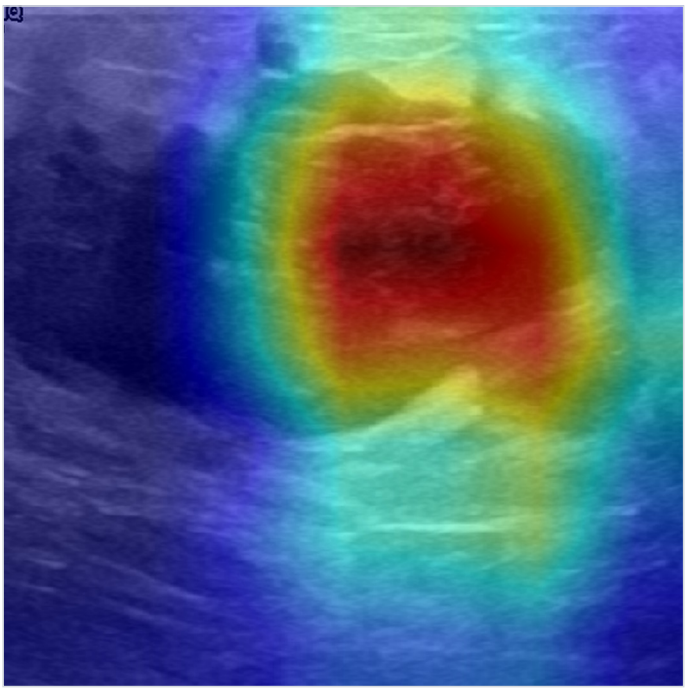


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Noura S.	Examination	Breast ultrasound
Age range	60–65	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

BENIGN

MODEL CONFIDENCE PER CLASS

Normal 2%

Benign 56%

Malignant 43%

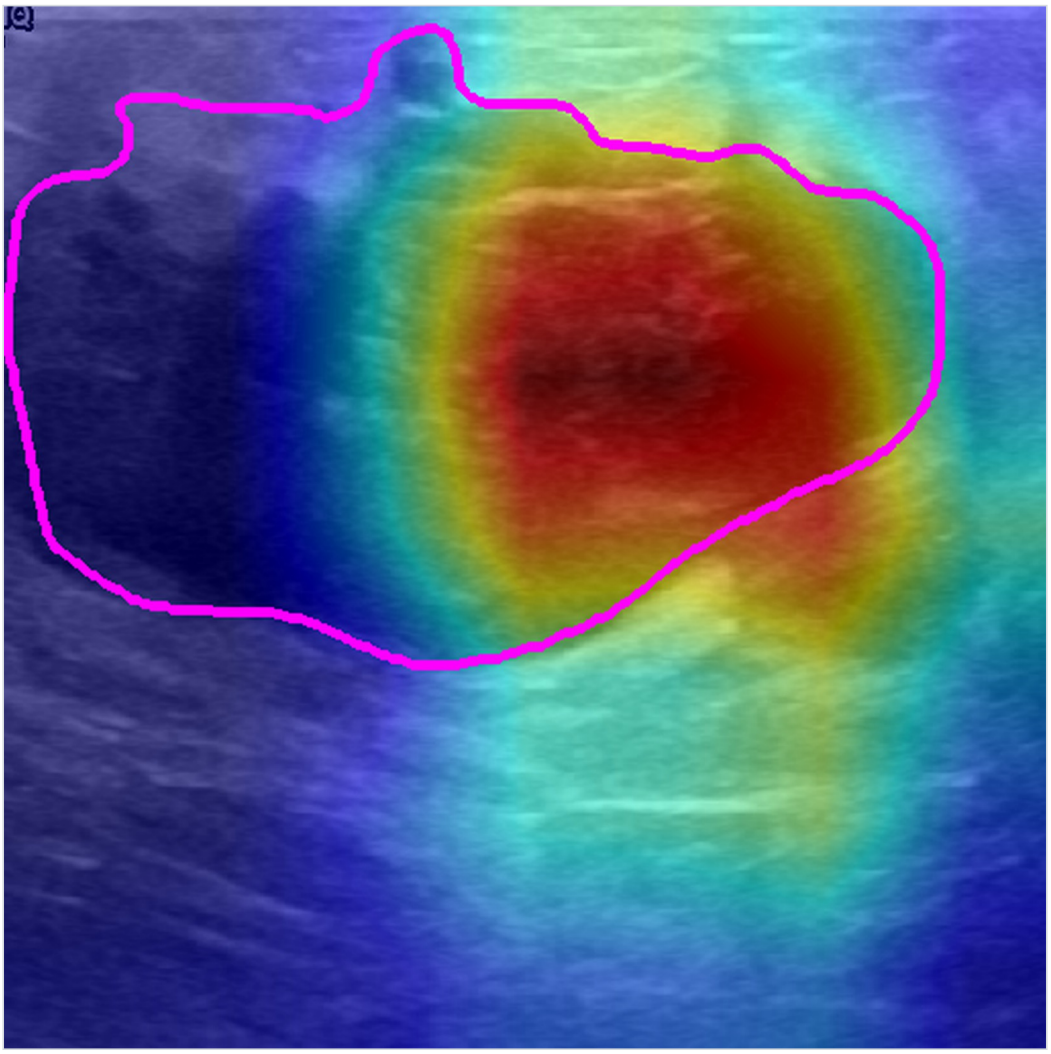
RECOMMENDATION

Likely benign. Routine follow-up suggested.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: MALIGNANT (cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The tumor is the large dark area. This time the AI's heatmap is right on it. The AI looked at the right place, then still said benign (56% benign vs 43% malignant).

Verdict

Folder 3: Looked right, judged wrong

LOOKED RIGHT, JUDGED WRONG

نظر إلى المكان الصحيح، وأخطأ الحكم

Arabic: draft, needs native review

The AI found the tumor, then misjudged it.

WHAT THIS TEACHES US

- Seeing isn't understanding: looking at the right spot is not enough.
- A heatmap on the tumor can make a wrong answer look trustworthy.
- 56% vs 43% is a close call, but the report only says "likely benign".
- For the patient it is still a missed cancer: false reassurance and delay.

How a human doctor protects the patient

- Judges the mass itself: its shape, edges and what lies around it.
- Treats a borderline result as a reason to look closer, not to relax.
- Combines the scan with other tests, such as a mammogram.
- Orders a biopsy when in doubt.

Arabic: draft, needs native review

- يحكم على الكتلة نفسها: شكلها وحوافها وما حولها.
- يعتبر النتيجة المتقاربة سببا للتدقيق أكثر، لا للاطمئنان.
- يجمع بين الصورة وفحوص أخرى، مثل صورة الثدي الشعاعية (الماموغرام).
- يطلب خزعة عند الشك.

QUESTION FOR YOU

The AI looked at the right spot but still got it wrong. Does a heatmap prove the AI understood?

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Breast Imaging · Ultrasound

PATIENT FOLDER

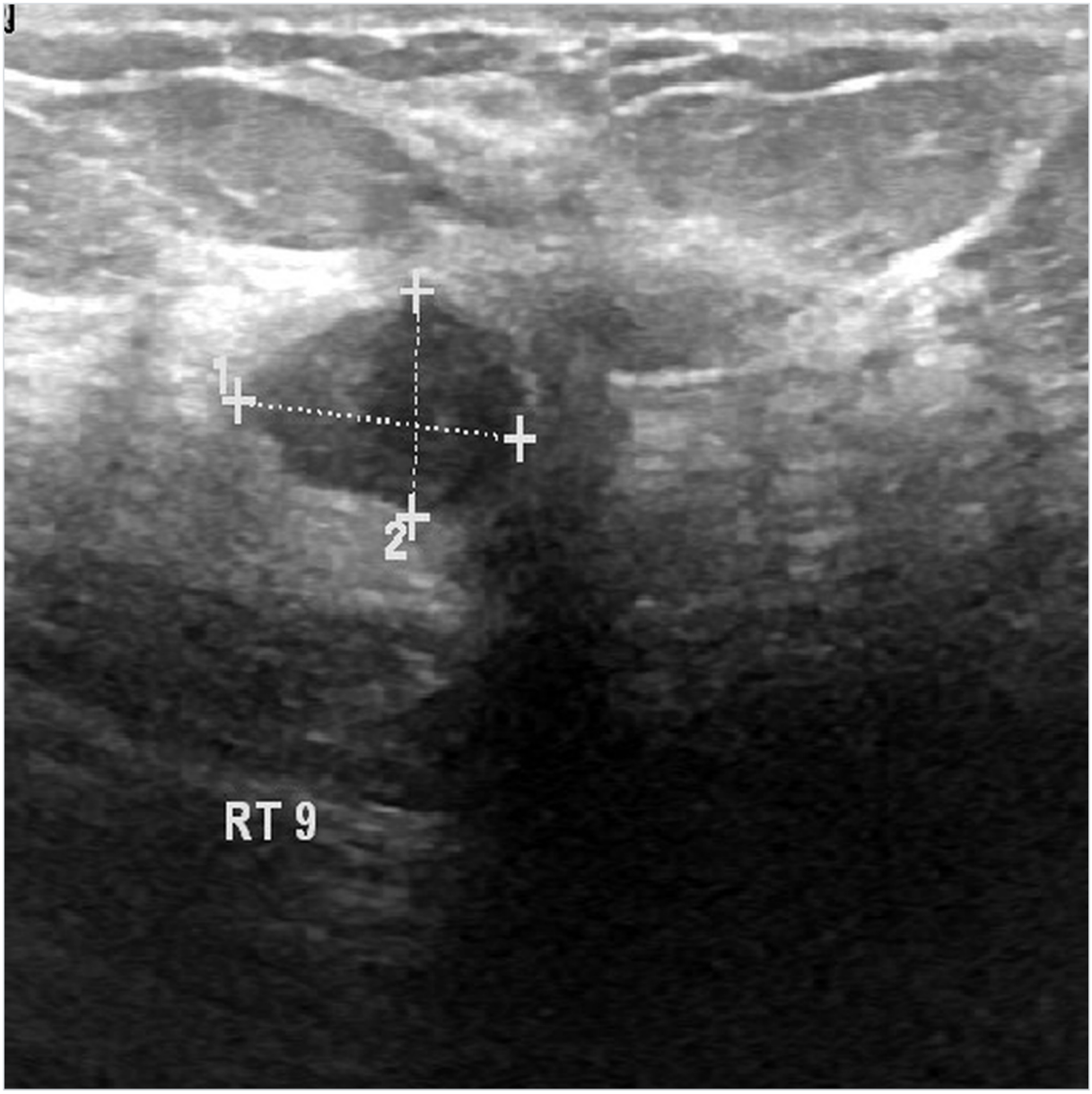
Folder 4

PATIENT	Aisha R.
AGE RANGE	30–35
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Follow-up of a lump noticed by the patient

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

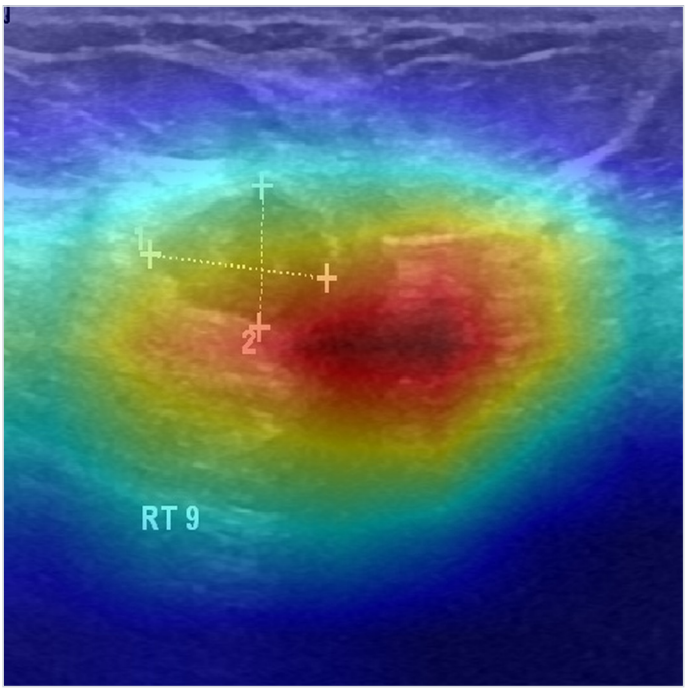


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Aisha R.	Examination	Breast ultrasound
Age range	30–35	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

MALIGNANT

MODEL CONFIDENCE PER CLASS

Normal	2%
Benign	22%
Malignant	**77%**

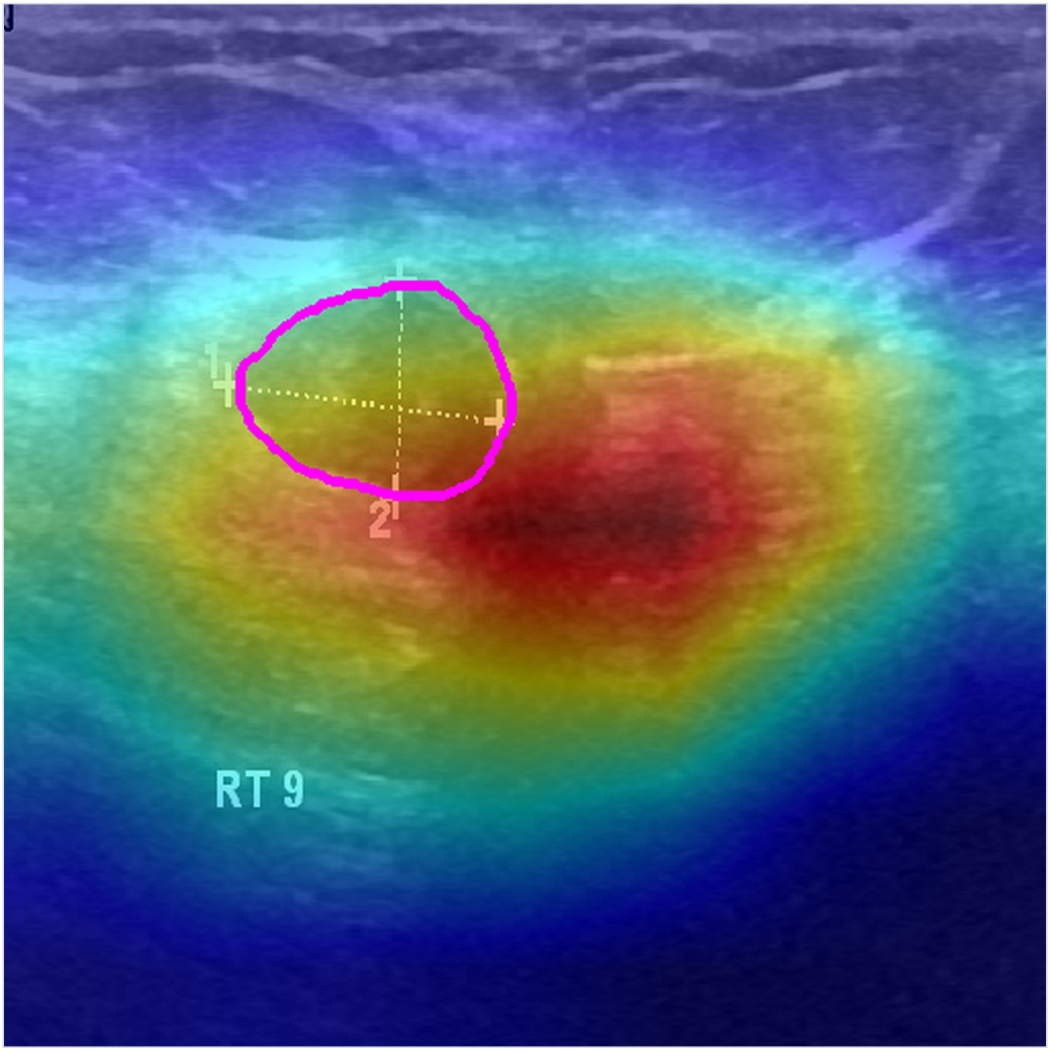
RECOMMENDATION

Suspicious for malignancy. Biopsy recommended.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: BENIGN (not cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The lump is the dark oval inside the pink line. The white "+" marks and numbers are measurement marks added by the person who scanned it. The AI's hottest area is beside the lump, and its warm area also covers some of those marks.

Verdict

Folder 4: False alarm

FALSE POSITIVE

نتيجة إيجابية كاذبة

Arabic: draft, needs native review

The AI raised the alarm. It was not cancer.

WHAT THIS COULD MEAN FOR THE PATIENT

- Fear and anxiety for the patient and family.
- Extra scans and an unnecessary biopsy.
- Cost and lost time.
- Busier clinics: every false alarm takes time from other patients.

How a human doctor protects the patient

- Compares with previous scans to see whether anything has changed.
- Looks at the lump's features, not just the software's score.
- A second reader can avoid an unnecessary biopsy.
- Explains the result calmly and clearly to the patient.

Arabic: draft, needs native review

- يقارن بالصور السابقة ليرى إن كان شيء قد تغير.
- ينظر إلى خصائص الكتلة، لا إلى درجة البرنامج فقط.
- يمكن لقارئ ثان أن يجنب المريضة خزعة غير ضرورية.
- يشرح النتيجة للمريضة بهدوء ووضوح.

QUESTION FOR YOU

Nothing was missed here. Is a false alarm harmless? Who pays the cost?

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PATIENT FOLDER

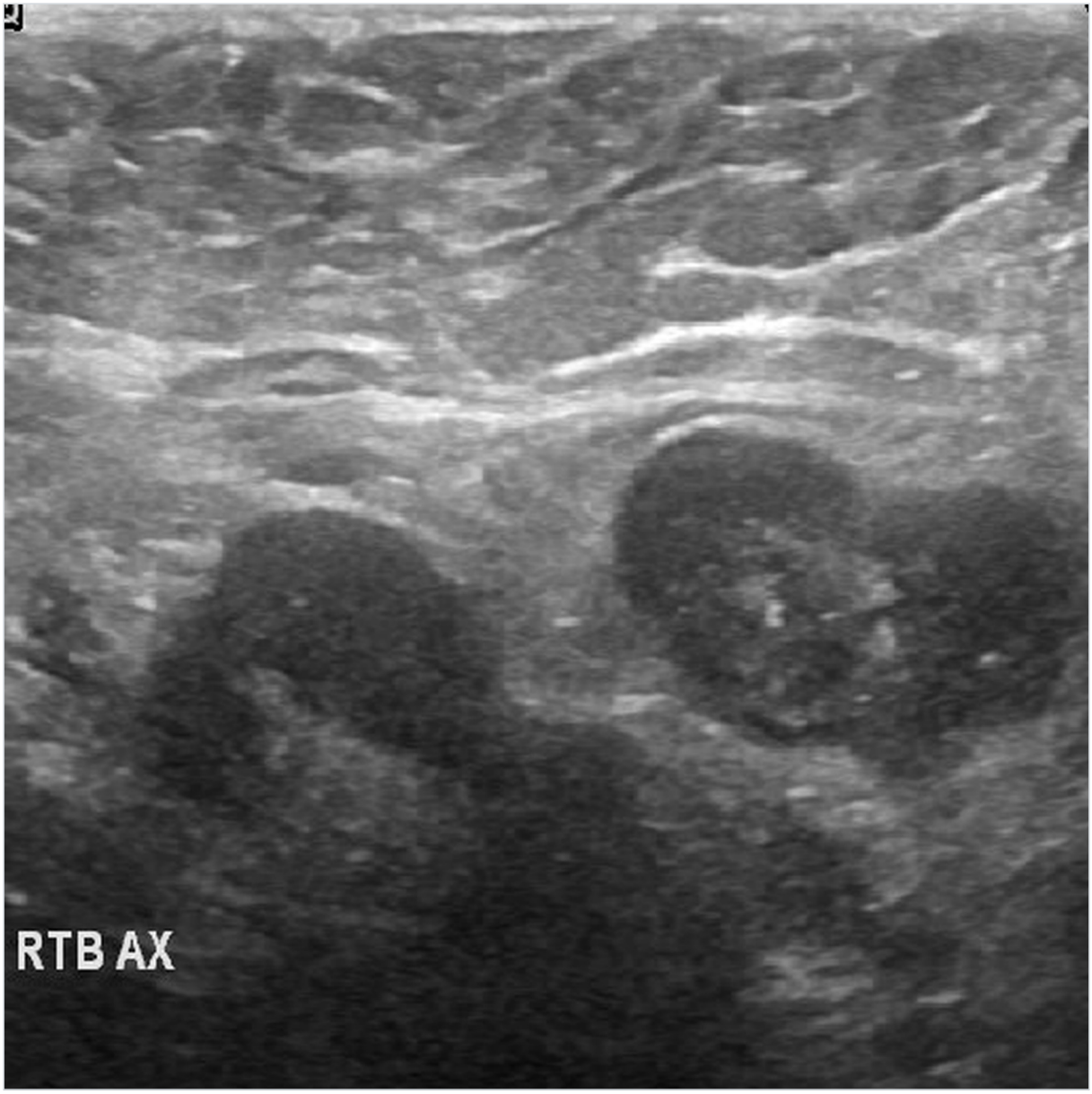
Folder 5

PATIENT	Hessa M.
AGE RANGE	50–55
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Further imaging after a change in breast shape

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

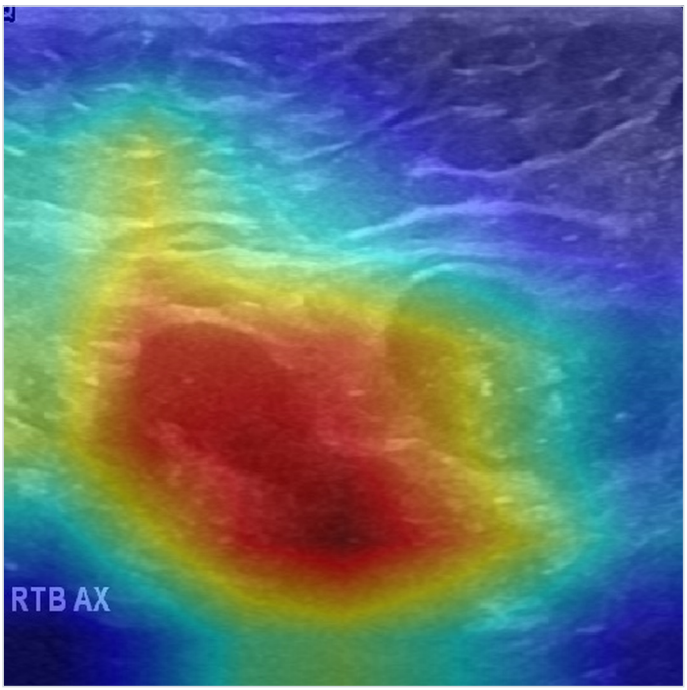


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Hessa M.	Examination	Breast ultrasound
Age range	50–55	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

MALIGNANT

MODEL CONFIDENCE PER CLASS

Normal 1%

Benign 10%

Malignant 89%

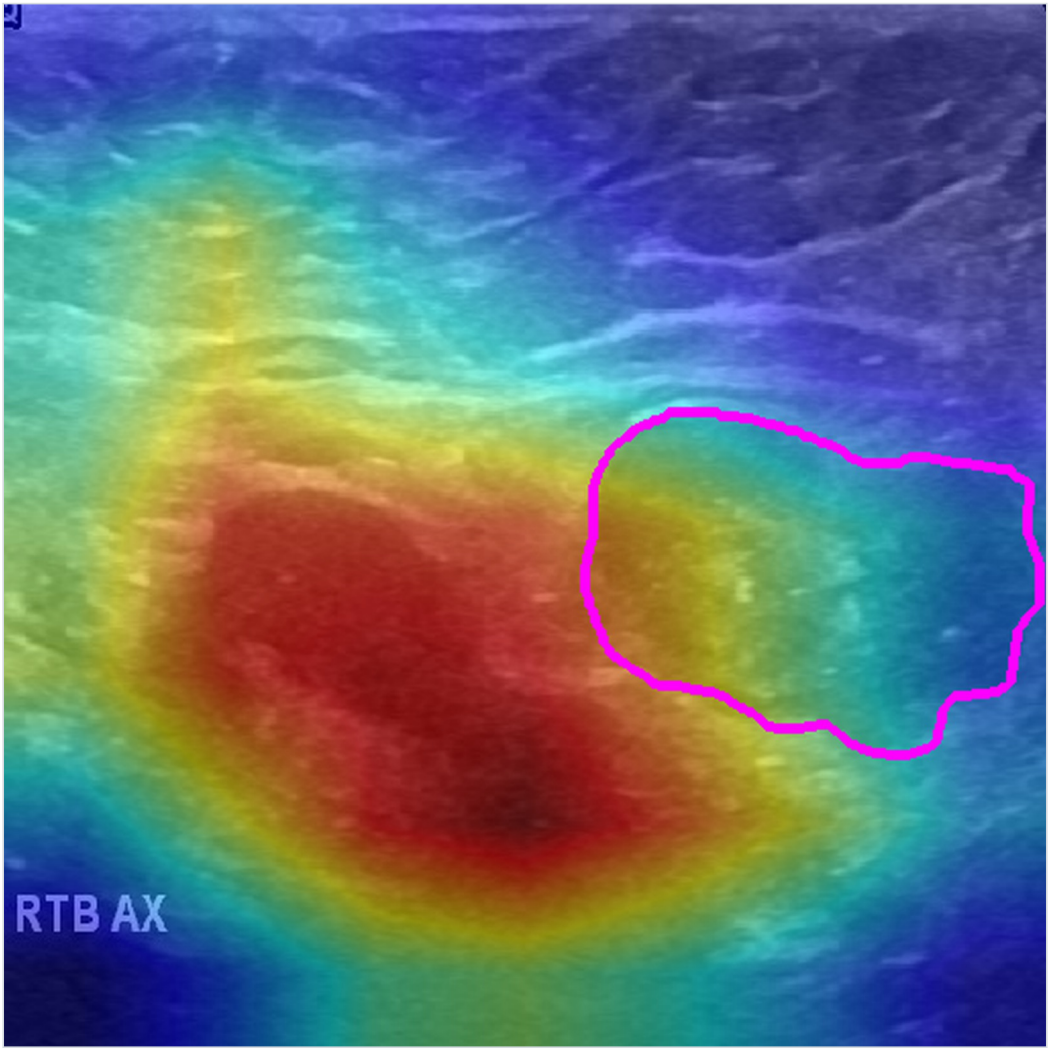
RECOMMENDATION

Suspicious for malignancy. Biopsy recommended.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: MALIGNANT (cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The tumor is inside the pink line on the right. The AI's heatmap is hottest on a different dark area to the left. The answer was right, but the AI was mostly looking somewhere else.

Verdict

Folder 5: Right answer, wrong reason

RIGHT ANSWER, WRONG REASON

إجابة صحيحة، لسبب خاطئ

Arabic: draft, needs native review

Correct (89% malignant), but mostly looking away from the tumor.

WHAT THIS TEACHES US

- Being right doesn't mean reasoning well.
- An AI that is right for the wrong reason can fail on the next patient.
- Checking only whether answers are right can hide this problem.
- This is why we look at heatmaps, and why we still need experts.

How a human doctor protects the patient

- Checks where the finding really is, not just the final label.
- A second reader reviews the images.
- Confirms the diagnosis with a biopsy.
- Looks at the whole patient before deciding on next steps.

Arabic: draft, needs native review

- يتحقق من مكان العلامة الحقيقي، لا من التصنيف النهائي فقط.
- يراجع قارئ ثان الصور.
- يؤكد التشخيص بالخزعة.
- ينظر إلى المريضة ككل قبل تحديد الخطوات التالية.

QUESTION FOR YOU

The AI got the right answer. Should we still worry about where it looked?

Hope Fair Teaching Clinic

Breast Imaging · Ultrasound

PATIENT FOLDER

Folder 6

PATIENT	Layla H.
AGE RANGE	65–70
EXAMINATION	Breast ultrasound
REASON FOR SCAN	Scan of a lump noticed during self-examination

FICTIONAL CASE — FOR EDUCATION ONLY

Patient details are invented. The ultrasound image is real and anonymous, from a public research dataset.

The scan

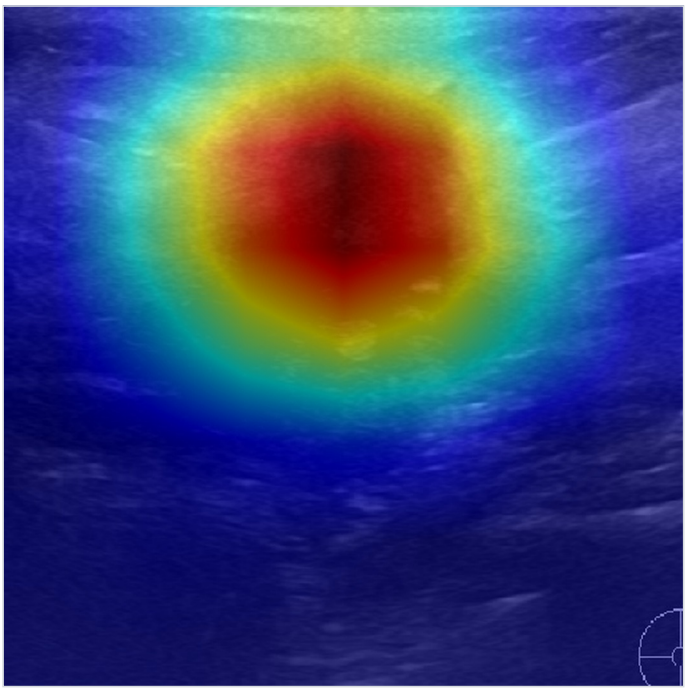


Breast ultrasound image. Look at it yourself before you turn the page.

AI analysis report

Patient	Layla H.	Examination	Breast ultrasound
Age range	65–70	Report type	Automated AI analysis

HEATMAP: WHERE THE AI LOOKED



AI PREDICTION

MALIGNANT

MODEL CONFIDENCE PER CLASS

Normal 1%

Benign <1%

Malignant 99%

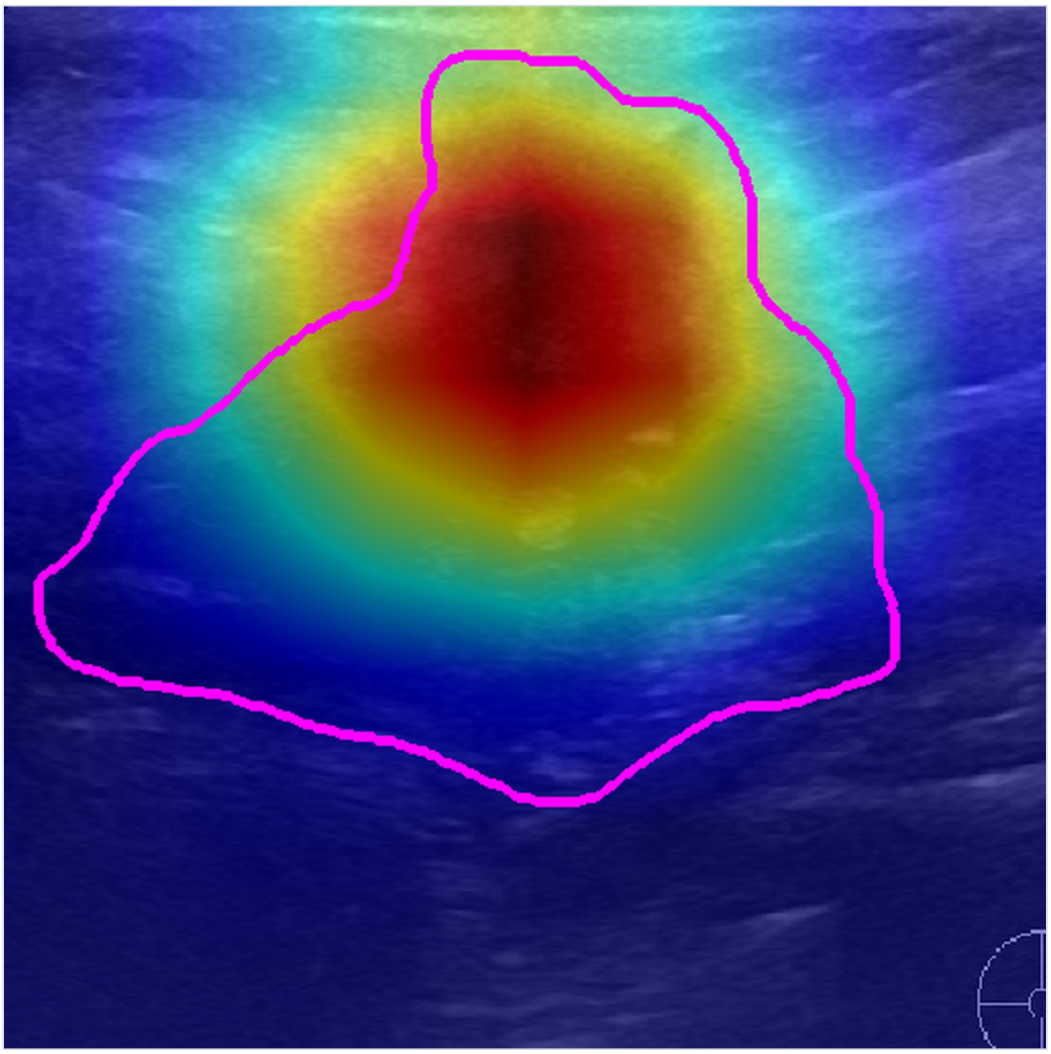
RECOMMENDATION

Suspicious for malignancy. Biopsy recommended.

Generated by an educational AI model. Not a medical device.

The real result

Biopsy showed: MALIGNANT (cancer).



Pink line = the real tumor outline, from the dataset. Colours = where the AI looked.

The tumor is the dark area inside the pink line. The AI's heatmap sits on the tumor and the answer was right. This is what helpful AI looks like.

Verdict

Folder 6: AI got it right

AI GOT IT RIGHT

أصاب الذكاء الاصطناعي

Arabic: draft, needs native review

Right answer, right place: 99% malignant, heatmap on the tumor.

WHAT THIS TEACHES US

- AI can help: it flagged the cancer and pointed to the right area.
- A highlight like this can help doctors decide where to look first.
- Cancers found early are more likely to be treated successfully.
- Even here, the biopsy, not the AI, confirmed the diagnosis.

How a human doctor protects the patient

- Uses the AI's highlight as a second opinion, not the final word.
- Confirms with a biopsy.
- Looks at the whole patient and talks through next steps with her.

Arabic: draft, needs native review

- يستخدم إشارة الذكاء الاصطناعي كرأي ثان، لا ككلمة أخيرة.
- يؤكد التشخيص بالخزعة.
- ينظر إلى المريضة ككل ويناقش معها الخطوات التالية.

QUESTION FOR YOU

When the AI is right, does the doctor still need to check? Why?